

Silicone based polyurea antifouling coatings modified by polyethylene glycol with enhanced wear resistance

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ARTICLE INFO

Keywords:

Marine antifouling
Polyaspartic ester polyurea
Silicone modification
Abrasion resistance

ABSTRACT

Polyurea coatings have attracted widespread attention due to their convenient application, excellent mechanical properties, and ability to meet the cleaning-resistance requirements of ships. However, their performance in inhibiting biological adhesion still needs improvement. Silicone-based materials have gained interest in marine antifouling owing to low surface energy, which is much lower than the adhesion of most marine organisms and environmental friendliness, but their mechanical properties often fall short of the requirements for cleaning resistance. In this work, a modified polyethylene glycol was synthesized via a thiol-ene click reaction. Subsequently, a polyurethane prepolymer was prepared, and aminopropyl silicone oil and polyaspartate were used as curing agents to produce a silicone-based polyurea coating modified with polyethylene glycol (IPEAx), combining the advantages of both polyurethane and silicone. The modified silicone-based polyurea coatings exhibited excellent antifouling properties for *Halophora yongxingensis* (*Halophora* sp.): after 7 days' immersion, the diatom adhesion inhibition rate of the IPEA3 coating reached $98.6 \pm 0.4\%$. Although the anti-adhesion effect on *Escherichia coli* (*E. coli*), *Staphylococcus aureus* (*S. aureus*), *Pseudoalteromonas xiamenensis* (*P. xiamenensis*) and *Nitzschia closterium* (*N. closterium*) was limited, but the adhesion strength, tensile strength, elongation and wear resistance increased by 25.63 %, 80.52 %, 69.47 % and 12.14 %, respectively. This strategy is expected to reduce the attachment of biofouling with both abrasion and cleaning resistance.

1. Introduction

Marine fouling organisms generally refer to animals, plants, and microorganisms that attach to the hulls of ships and the surfaces of installations in the sea [1,2]. These organisms impact human activities at sea and restrict the exploration of marine resources [3,4]. Currently, there are more than 5000 known species of marine fouling organisms [5], including bacteria, diatoms, mussels, barnacles, and others. The overgrowth of fouling organisms attached to hull surfaces and leads to several negative effects [6], such as accelerated substrate corrosion and reduced ship service life; increased navigational resistance, which endangers ship safety; higher fuel consumption during navigation; and consequently increased carbon dioxide emissions, exacerbating the greenhouse effect. In addition, the biological invasion caused by fouling organisms adhering to hulls has become increasingly serious, severely

disrupting local ecosystems [7,8].

Various strategies have been proposed to address the challenges of hull fouling, including mechanical cleaning [9], laser irradiation [10] and the application of antifouling coatings [11–13]. Among these, antifouling coatings are extensively employed due to their high efficiency and relatively low cost [14,15], and they remain the most effective method for combating fouling on hull surfaces. The outstanding antifouling properties of tributyltin-based coatings were once widely commercialized [16], but it was later discovered that tributyltin release severely damaged marine environments and threatened marine life [17]. In recent years, research efforts have increasingly focused on the exploration of new marine antifouling coatings, for instance, self-polishing antifouling coatings [18,19], bionic antifouling coatings [20,21], and fouling release antifouling coatings [22–24]. Self-polishing antifouling coatings have prominent advantages in antifouling

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efficiency and economic benefits, but environmental issues remain a challenge. The future trend is to develop more environmentally friendly and adaptable antifouling technologies to balance performance and ecological needs [25,26]. The commercialization of bionic antifouling coatings faces multiple challenges, including high costs and low efficiency, limited to laboratory research and difficult to be applied in engineering. Fouling release antifouling coatings have the best environmental friendliness, but their static antifouling performance is poor. Fluoropolymers such as polytetrafluoroethylene (PTFE) and polyvinylidene fluoride (PVDF) have extremely low surface energy of 15–18 mN/m, which can be used to prepare fouling release antifouling coatings [27–29]. However, their cost is relatively high, and with the restriction on the use of fluorides, fluorine-free antifouling coatings have become the mainstream. Thanks to the extremely low surface energy and good flexibility, polydimethylsiloxane (PDMS) is the most commonly used matrix material for fouling release antifouling coatings [30–33]. Studies have shown that the surface energy of PDMS coatings is usually 20–24 mN/m, which is much lower than the adhesion of most marine organisms [34]. Nevertheless, silicone materials have deficiencies in mechanical strength, adhesion, and wear resistance. Current advancements in antifouling coatings include desorption-type coatings engineered through interface property design. These coatings feature a unique mechanism where contaminants adhering to the surface can be mechanically removed, allowing the cleaned coating to be reused and extending its service life.

Polyurethane (polyurea) materials are widely used in coatings [35], adhesives [36,37], and the automotive industry [38,39] due to their excellent solvent resistance, strength, and impact resistance. Polyurethane (polyurea) contains abundant hydrogen bonds and exhibits microphase separation, endowing it with outstanding mechanical properties, making it attractive for marine antifouling applications. However, polyurethane itself lacks inherent antifouling properties and thus requires modification. Xie et al. [40] reported a nanodiamond particles modified by perfluoroalkyl groups based on PDMS-PUa, via this strategy, the tensile strength and adhesion were increased by 33 % and 46 %, respectively. Antifouling tests showed that the coatings maintained antifouling performance after four months' immersion in natural seawater. A reproducible self-healing composite coating via epoxy-modified polyurethane (FD-EPU) was reported by Tian et al. [41]. Functionalized reduced graphene oxide imparted FD-EPU with excellent antifouling properties, achieving over 97.5 % bacteria removal and up to 96.2 % microalgae removal. Hong et al. [42] reported an original industrial waterborne polyurethane (WPU) antifouling coating (FSiWPU) by introducing hydroxyl-terminated siloxane fluoride (HTFSi) via a facile polycondensation reaction into a polyether-type WPU. Galhenage et al. [43] evaluated the real-world performance of siloxane-polyurethane (SiPU) coatings under freshwater conditions, and two compositions showed excellent field performance for up to two years.

Although antifouling coatings are an effective means of preventing marine fouling, current coatings still have limitations. Over time, fouling organisms inevitably adhere to hull surfaces, necessitating cleaning. During high-pressure washing or mechanical cleaning, it is crucial that the coating remains undamaged and continues to exhibit antifouling performance [44]. Therefore, developing antifouling coatings with both abrasion and cleaning resistance is of significant importance to decrease the hazard of alien biological incursion. Wu et al. [45] developed an innovative green antifouling and wear-resistant polyurethane coating based on lignin, the mechanical strength was enhanced by approximately 200 %, furthermore, compared with commercial coatings, lignin-containing coatings shown superiority in biofilm suppression of both model marine bacteria and the biofilm-model strain *P. aeruginosa*. Han et al. [46] integrating PDMS as the low-surface-energy component and immobilizing it within the polyurea matrix through interpenetrating polymer network (IPN) technology, the coating benefits from both materials, exhibited excellent overall performance in terms of mechanical properties, anti-fouling resistance, chemical resistance and abrasion

resistance. An et al. [47] developed eco-friendly and high-efficiency modified silicone-based fouling release coatings with robust mechanical properties and sustainable marine antifouling performance, the mechanical strength and substrate adhesion was enhanced to 6.4 MPa and 2.87 MPa, respectively.

Silicone exhibits excellent dynamic antifouling properties due to their low surface energy, but their mechanical performance can't meet the requirements for durable cleaning-resistant coatings. Polyurea possess sufficient mechanical strength to satisfy cleaning durability demands, furthermore, polyaspartic acid ester-based polyurea has gained widespread attention for its superior wear resistance, however, active antifouling capability remains underdeveloped. Hence, in this work, polyurethane prepolymers, aminopropyl silicone oil, and polyaspartate were used as curing agents to prepare a polyethylene glycol-modified silicone-based antifouling coating (IPEAx). Hardness, adhesion, and tensile strength were also tested to evaluate the mechanical properties of coatings. Mass loss and water contact angle were recorded during wear resistance testing with varying numbers of friction cycles. Finally, bacterial and diatom attachment experiments as well as mussel settlement tests were investigated to assess the antifouling performance of the coatings.

2. Experiment

2.1. Materials

β -mercaptoethanol (MCE), polyethylene glycol methyl ether methacrylate (PEGMA), and 2,2-dimethoxy-2-phenylacetophenone (DMPA) were obtained from Sigma-Aldrich. Tetrahydrofuran was purchased from by Tianjin Damao Chemical Co., Ltd. Polypropylene glycol (PPG) and dibutyltin dilaurate (DBTDL) were obtained from Macklin, while isophorone diisocyanate (IPDI) was purchased from Union Wanhua. Aminopropyl silicone oil was sourced from Suzhou Cheerchem Advanced Material Co., Ltd., and polyaspartate ester resin (NH1420) was supplied by Desmophen.

2.2. Preparation of modified polyethylene glycol polyurethane prepolymer (IPE)

The modified polyethylene glycol (PM) was synthesized by reacting β -mercaptoethanol (MCE) with polyethylene glycol methyl ether methacrylate (PEGMA) via a thiol-ene click reaction under UV irradiation. The preparation process is illustrated in Fig. 1.

Briefly, 28 g of PEGMA was dissolved in tetrahydrofuran, followed by the addition of 3.45 g of MCE with stirring. Next, 0.94 g of DMPA was added as a photoinitiator. The molar ratio of MCE to PEGMA was 1.5:1 to ensure complete reaction of the double bonds on PEGMA and successful grafting. In order to maintain an inert atmosphere, nitrogen was purged for 10 min to exhaust dissolved oxygen. UV irradiation was then applied while shielding from ambient light. After 3 h of stirring, a clear, light-yellow transparent solution was obtained. The excess MCE was removed by washing with deionized water, yielding the final product referred to as PM.

Next, 90 g of PPG was placed in a three-necked flask and dehydrated at 105 °C under vacuum for 2 h. After cooling the system to below 45 °C, PM was added under stirring, followed by the slow addition of 87.1 g of IPDI, while keeping the temperature below 50 °C. Then, DBTDL of 0.11 g was added as a catalyst, and heated to 80 °C and maintained for 3 h to obtain the isocyanate-terminated graft-modified polyethylene glycol polyurethane (IPE) prepolymer.

2.3. Preparation of polyethylene glycol modified silicone polyurea coating (IPEAx)

Typically, isocyanate-sealed IPE prepolymer of 10 g and NH1420 of 3.94 g were mixed evenly, exhaust the bubbles via vacuum, and then

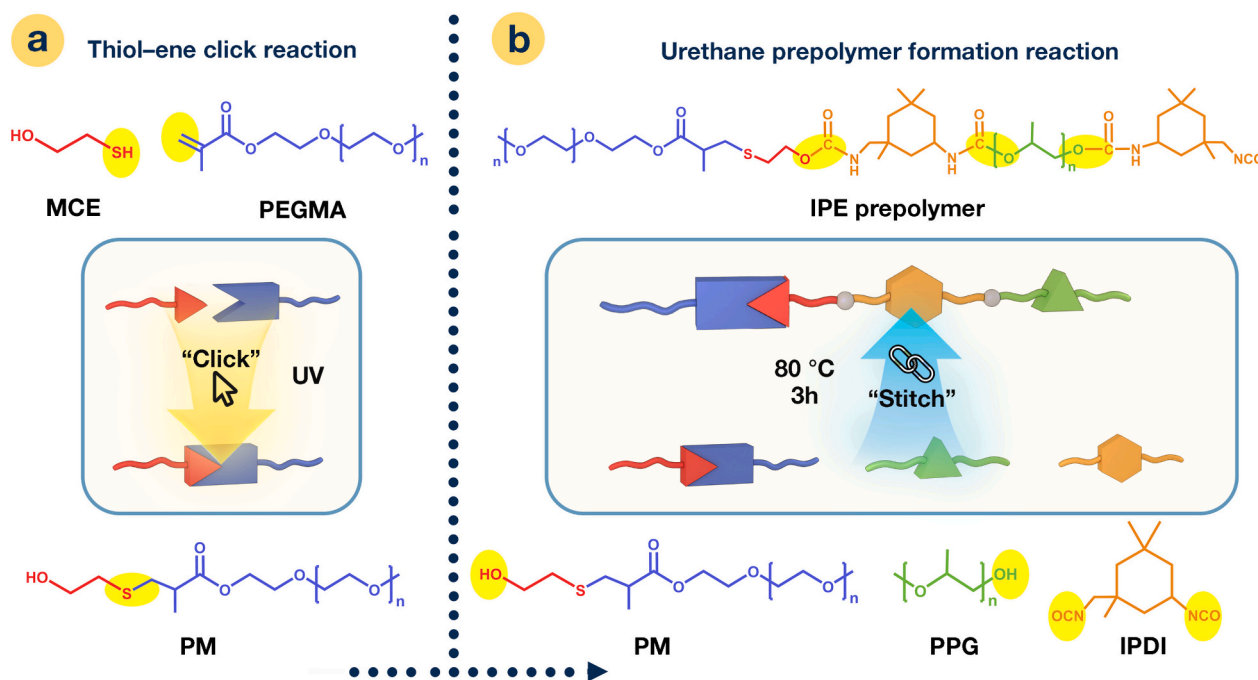


Fig. 1. Schematic of (a) PM and (b) IPE prepolymer preparation.

coated on the surface of glass (76 mm × 26 mm × 1 mm) or Q235 carbon steel (100 mm × 150 mm × 3 mm) by spinning, curing at 80 °C for 12 h, IPE coating was obtained. For the preparation of the polyethylene glycol-modified silicone aspartate polyurea coatings (IPEAx), aminopropyl silicone oil was incorporated as part of the curing agent. The molar ratios of aminopropyl silicone oil to PEGMA were adjusted to 1:1, 2:1, and 3:1, resulting in coatings designated as IPEA1, IPEA2, and IPEA3, respectively.

The remaining isocyanate groups were cured with NH1420, following the same procedure used for the IPE coating. The composition details of the coatings are provided in Table 1.

3. Results and discussion

3.1. Characterization of coatings

Fig. 2(a) are the FTIR spectra of PEGMA, MCE, and PM. For PEGMA, the absorption peak observed at 1640 cm^{-1} is attributed to the C=C bond stretching vibration [48], while for MCE, the peak at 2550 cm^{-1} is attributed to the -SH group [49]. In the spectrum of PM, the characteristic peaks for C=C and -SH bonds disappeared, indicating complete reaction between MCE and PEGMA. Furthermore, the appearance of a strong -OH stretching vibration peak at 3473 cm^{-1} confirms the composition of PM. The FTIR spectrum of the IPE prepolymer is shown in Fig. 2(b). Peaks at 1530 cm^{-1} and 3350 cm^{-1} are attributed to the bending and stretching vibrations of the N-H group, respectively. The peak at 1718 cm^{-1} corresponds to the C=O bond in the carbamate group (-NH-C=O), this was ascribed to the reaction of isocyanate groups and hydroxyl groups. The absence of the -OH characteristic peak indicates that the hydroxyl groups in both PPG and PM have completely

reacted with the isocyanate groups. Additionally, the presence of a characteristic -NCO peak at 2267 cm^{-1} confirms the successful synthesis of the isocyanate-terminated IPE prepolymer.

Fig. 2(c) are the ATR-FTIR spectra of the IPE and IPEAx coatings. For the IPEAx coatings, characteristic peaks at 1259 cm^{-1} and 806 cm^{-1} correspond to the bending and stretching vibrations of the Si-CH₃ groups, respectively. The peak at 3365 cm^{-1} is attributed to the N-H stretching vibration. The absence of the -NCO peak near 2270 cm^{-1} suggests exhaustive reaction of the IPE prepolymer with the curing agents, indicating full curing of the coatings.

Fig. 2(d) is the XPS survey spectrum of IPEA1, revealing the presence of C, N, O, Si, and S elements, confirming the composition of the coating. As shown in Fig. 2(e), the characteristic peaks at 102.5 eV and 101.9 eV were assigned to Si-O and Si-C bonds, respectively, indicating successful grafting of aminopropyl silicone oil. In Fig. 2(f), the peaks at 289.2, 286.5, 285.1, and 284.6 eV correspond to C=O, C-O/C-N/C-S, C-C, and C-Si bonds, respectively. These results demonstrated the IPEA1 coating meet expectation.

Various factors will lead to coating failure, with thermal degradation being one of the key causes. The thermal degradation curve of the coatings was tested through thermogravimetric analysis, and the effect of aminopropyl silicone oil on thermal stability was assessed. The TGA curves and weight loss rates of the coatings are shown in Figs. 2(g) and 2(h). Thermal stability was evaluated based on the temperature corresponding to 5 % total mass loss (T_{d5}) and the temperature of maximum thermal decomposition rate (T_{max}) [50]. The T_{d5} values for IPE, IPEA1, IPEA2, and IPEA3 were 277.36 °C, 279.16 °C, 287.49 °C, and 287.36 °C, respectively. The T_{max} values were 395.17 °C, 400.32 °C, 429.17 °C, and 428.69 °C, respectively. Improving thermal stability can improve the reliability of coating. After the addition of aminopropyl silicone oil, the T_{d5} of the IPEAx coatings initially increased and then stabilized. This enhancement can be ascribed to the increased cross-linking density provided by the aminopropyl silicone oil, which enhances thermal resistance. Once an equilibrium in cross-link density is reached, further improvements plateau, resulting in minimal changes in T_{d5} , with IPEA3 achieving 287.36 °C. The trend in T_{max} was consistent with that of T_{d5} . The T_{max} of the IPE coating was 395.17 °C, while T_{max} values for the IPEAx coatings increased to more than 400 °C, this is the contribution of

Table 1

Raw and processed material of IPE and IPEAx coating.

Sample	IPE prepolymer/g	Aminopropyl silicone oil/g	NH1420/g
IPE	10	0	3.94
IPEA1	10	1.14	3.34
IPEA2	10	2.28	2.78
IPEA3	10	3.41	2.27

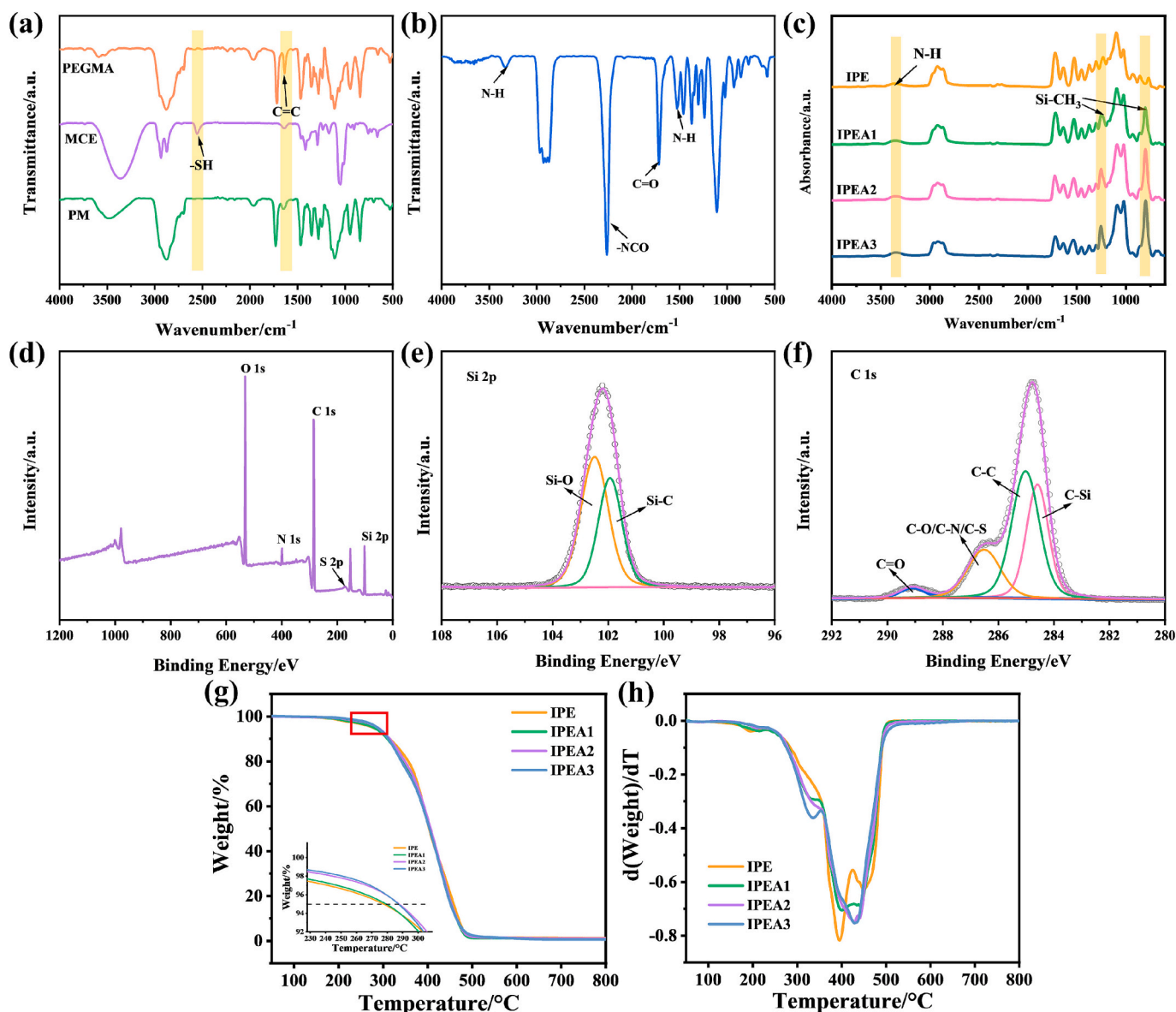


Fig. 2. FT-IR spectra of (a) PEGMA, MCE, PM and (b) IPE prepolymer. (c) ATR-FTIR spectrogram of coatings, (d) XPS full spectra of IPEA1, (e) Si 2p high resolution spectra of IPEA1, (f) C 1 s high resolution spectra of IPEA1, (g) TGA of coatings, (h) mass loss rate of coatings.

the soft segments decompose in the polymer chains. Additionally, all coatings exhibited a decomposition peak around 330 °C before reaching T_{max} , which is associated with the breakdown of urea and carbamate bonds [51].

The SEM was used to observe the microscopic morphology of the coatings. Figs. 3(a–d) shown the SEM images of IPE, IPEA1, IPEA2, and IPEA3. The results show that the coating surfaces are flat and smooth. Surface roughness was further evaluated by AFM. As shown in Figs. 3(e–h), R_a of IPE, IPEA1, IPEA2, and IPEA3 were 0.256, 0.263, 0.259, 0.274 nm, respectively, and R_q were 0.321, 0.329, 0.325, 0.345 nm, respectively. The surface roughness remained consistent across all samples, indicating that the supplement of aminopropyl silicone oil did not affect the surface morphology of the coatings.

The antifouling efficacy of coatings is affected by hydrophilic and hydrophobic properties of surface. Therefore, the static water contact angle of each coating was measured, as shown in Fig. 3(i). Due to the presence of hydrophilic polyethylene glycol methyl ether methacrylate segments, the water contact angle of IPE coating was $78.3 \pm 0.4^\circ$, shown hydrophilicity. After the incorporation of aminopropyl silicone oil, the water contact angles of the IPEA1, IPEA2, and IPEA3 coatings increased

to $94.3 \pm 0.5^\circ$, $95.6 \pm 0.9^\circ$, and $98.1 \pm 0.8^\circ$, respectively. This trend shows that increasing the aminopropyl silicone oil content enhances the hydrophobicity of the coatings. Additionally, the contact angles with a non-polar solvent (n-hexadecane) were measured, as shown in Fig. 3(j). Usually, Owens-Wendt formula was used to calculate the surface energy of coating, water and n-hexadecane were polar and non-polar liquid, respectively. The surface energy of the IPE coating (Fig. 3(k)) was $35.11 \text{ mJ}\cdot\text{m}^{-2}$. After the addition of aminopropyl silicone oil, the surface energy of the IPEAx decreased to between 27 and $28 \text{ mJ}\cdot\text{m}^{-2}$, falling within the range typically associated with low surface energy coatings [52].

3.2. Mechanical behavior and wear resistance of coatings

Excellent mechanical performance was critical for extending the service life of coatings. In this study, the adhesion strength on Q235 carbon steel was evaluated by pull-off tests (ASTM D4541–2009), and Shore hardness and tensile properties were evaluated to discuss the effect of aminopropyl silicone oil on mechanical performance. The adhesion strength (Fig. 4(a)) of the IPE coating was $6.75 \pm 0.91 \text{ MPa}$. After

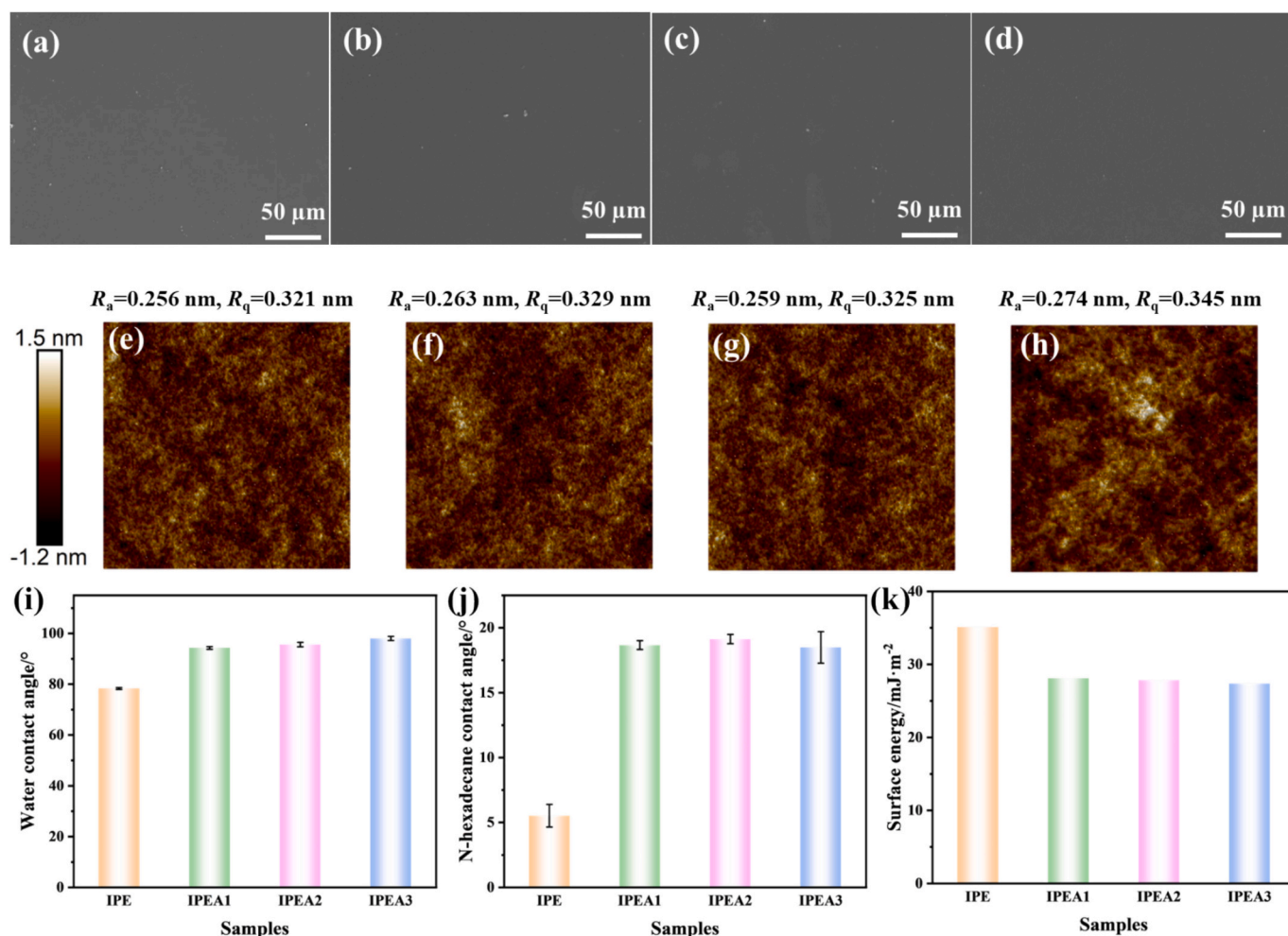


Fig. 3. SEM images of (a) IPE, (b) IPEA1, (c) IPEA2, (d) IPEA3. AFM images of (e) IPE, (f) IPEA1, (g) IPEA2, (h) IPEA3. (i) Water and (j) n-hexadecane contact angle of different coatings, (k) calculated surface energy of different coatings.

the incorporation of aminopropyl silicone oil, the adhesion strength increased significantly, this is the contribution of strong interaction between the coating and the substrate via $-NH_2$ and $-OH$ groups [53–55]. However, with further increases in aminopropyl silicone oil content, the adhesion strength shown a slight decrease trend. This may be the negative effect of increased coating flexibility. Nonetheless, the adhesion strength of the IPEA3 coating (8.48 ± 0.57 MPa) is remained higher than IPE. The Shore hardness results are presented in Fig. 4(b). With increasing aminopropyl silicone oil content, the Shore hardness of the coatings showed a downward trend. This can be owed to the enhanced flexibility of the polymer chains and increased cross-linking density, which collectively improve coating flexibility but slightly reduce hardness.

Fig. 4(c) shown the stress-strain curve of the coatings. The tensile strengths of IPE, IPEA1, IPEA2, and IPEA3 were 0.77 ± 0.23 , 2.39 ± 0.11 , 2.11 ± 0.38 , and 1.39 ± 0.25 MPa, respectively. The elongations of IPE, IPEA1, IPEA2, and IPEA3 were 476.21 ± 11.69 %, 546.17 ± 18.49 %, 688.39 ± 12.23 %, and 807.01 ± 10.58 %, respectively. Compared with IPE, the tensile strength was increased after the addition of aminopropyl silicone oil, this can be ascribed to the $-NCO$ group reacted with $-NH_2$ to form a stable urea bond ($-NH-CO-NH-$) that the cohesion was enhanced, continue to increase aminopropyl silicone oil, which plays a negative role in tensile strength, this can be attributed to the increased cross-linking points introduced by aminopropyl silicone oil. However, excess aminopropyl silicone oil enhanced the flexibility of the coating segments and decreased the tensile strength. The elongation

of the coatings enhanced in the order of IPE, IPEA1, IPEA2, and IPEA3, and the elongation of the IPEA3 coating reached 807.01 ± 10.58 %. This can be ascribed to the increased chain segments with more aminopropyl silicone oil, which enhanced the flexibility of the coating, thus increasing elongation. Other reported mechanical properties are shown in Table S1, these results demonstrated that the tensile strength, elongation, and adhesion strength were balanced in this work.

The mass loss of coatings (Fig. 4(d)) during the wear-resistance test was shown that with an increasing of friction cycles, the weight loss of the coatings decreased. Compared with the IPE coating, the mass loss of the IPEAx coatings was decreased after the addition of aminopropyl silicone oil, IPEA1 shows the best wear resistance. In the first, second, third, and fourth 200-cycle friction tests, the mass loss was 14.1 ± 0.1 mg, 13.5 ± 0.2 mg, 12.7 ± 0.5 mg, 12.7 ± 0.6 mg, respectively. This can be ascribed to the formation of urea bond ($-NH-CO-NH-$), which increased the cohesion of the coating and reduced the mass loss after friction. The mass loss increased slightly with further addition of aminopropyl silicone oil. This can be ascribed to enhanced flexibility. After the first 200 cycles of friction, the weight loss of the IPEA3 coating was 17.3 ± 0.6 mg. In the second, third, and fourth 200-cycle friction tests, the mass loss was 16.7 ± 0.4 , 15.0 ± 0.2 , and 14.1 ± 0.3 mg, respectively. The results show that when the softness and hardness were balanced, the wear resistance of coating was optimal. The water contact angle after every 200 cycles of friction was measured (Fig. 4(e)). Compared with the original coatings, the water contact angle increased after friction. As the increase of friction cycles, the water contact angle

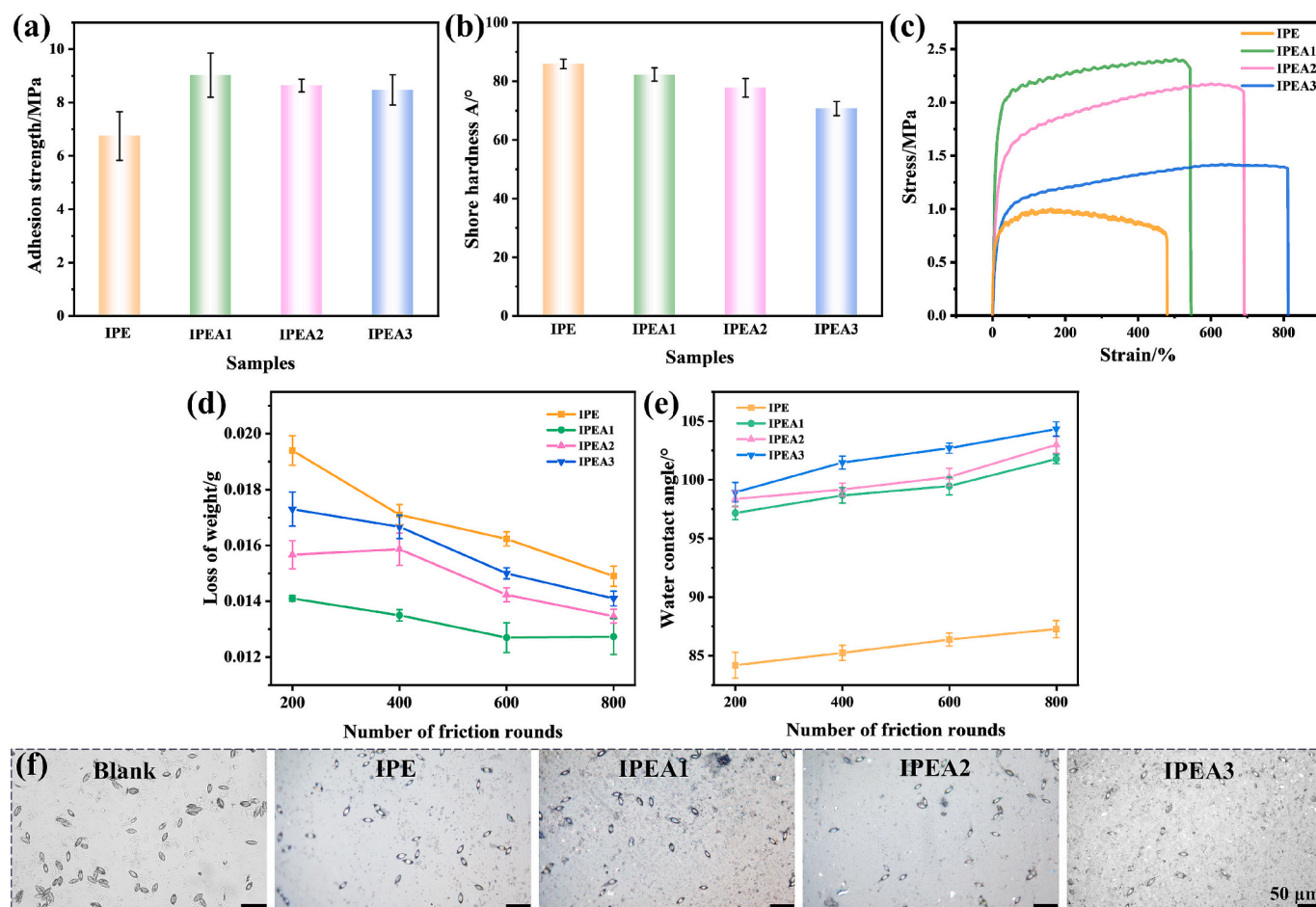


Fig. 4. (a) Adhesion strength of coatings, (b) Shore hardness of coating, (c) stress-strain curve of samples, (d) mass loss of wear-resistant test, (e) water contact angle of coatings after wear-resistant test, (f) optical microscope images of coating after 800 friction rounds immersed in *Halamphora* sp. solution for 1 day.

showed a gradual increase, this can be attributed to increase in surface roughness. Due to the hydrophobic aminopropyl silicone oil, the IPEA3 coating exhibited the highest water contact angle. After 800 cycles of friction, the coatings were immersed in an algal culture solution for 1 day. Compared with the control sample (Fig. 4(f)), the IPE and IPEAx coatings showed the ability to inhibit the adhesion of *Halamphora yongxingensis* (*Halamphora* sp.) after the friction experiment. The increased surface roughness enhanced the hydrophobicity of the coatings, this is benefit for inhibiting diatom adhesion for a short time.

3.3. Antibacterial performance of coatings

Fig. 5(a-c) and (d-f) show the photos of bacterial colonies on different coatings and the antibacterial ratio of different coatings, respectively. Both IPE and IPEAx coatings showed antibacterial properties. The antibacterial ratios of the IPE coating against *Escherichia coli* (*E. coli*), *Staphylococcus aureus* (*S. aureus*), *Pseudoalteromonas xiamenensis* (*P. xiamenensis*) were $61.1 \pm 1.3\%$, $49.5 \pm 1.7\%$, and $53.2 \pm 1.4\%$, respectively. The antibacterial property of the IPE coating was attributed to the hydrophilicity of the coating, which was prepared by modified polyethylene glycol methyl ether methacrylate. When the coating contacted the bacterial solution, a hydration layer reducing the adhesion of bacteria. The aminopropyl silicone oil of coating shown a positive effect for anti-bacterial property, with the increase of aminopropyl silicone oil, the anti-bacterial ratio was improved. The antibacterial property of IPEAx coatings was attributed to the low surface energy (27–28 mN/m), which makes it hard for bacteria to attach to the surface. The antibacterial ratios of the IPEA3 coating against *E. coli*,

S. aureus and *P. xiamenensis* were $61.4 \pm 2.3\%$, $58.2 \pm 1.2\%$, and $64.9 \pm 2.1\%$, respectively. Compared with IPE coating, the IPEAx shown a limited improvement in antibacterial performance. This can be ascribed to the IPE coating actively blocks bacterial adhesion through the hydration layer, after the addition of amino silicone oil, the surface energy of the cross-linking coating is reduced, and the bacterial adhesion is reduced by reducing the spread of water molecules on the surface, therefore, the passive repulsion effect is limited.

3.4. Anti-diatom performance of coatings

In this work, two typical diatoms were used to explore the anti-fouling properties of different samples. *Halamphora* sp. is a typical alga with strong adhesive ability, and *Nitzschia closterium* (*N. closterium*) is a phytoplankton commonly present in seawater. The blank sample was glass sheet. The glass sheet and the experimental coatings were immersed into algal culture solutions for different days. The attachment of diatoms on different coatings were shown in Fig. 6(a, b). A large number of diatoms was observed on glass sheet surface, while the number attached to the IPE coating was decreased notable, indicating that a hydration layer formed on the hydrophilic surface, inhibiting diatom adhesion. Furthermore, compared to control surfaces, the IPEAx coatings showed markedly fewer adhered diatoms, highlighting the inhibitory role of low surface energy. The anti-diatom ratios of different coatings are shown in Fig. 6(c, d). The inhibition rate of diatom attachment on IPE and IPEAx coatings decreased slightly with increasing immersion time. After 7 days of immersion, the inhibition rates of the IPE, IPEA1, IPEA2, and IPEA3 coatings on *Halamphora* sp.

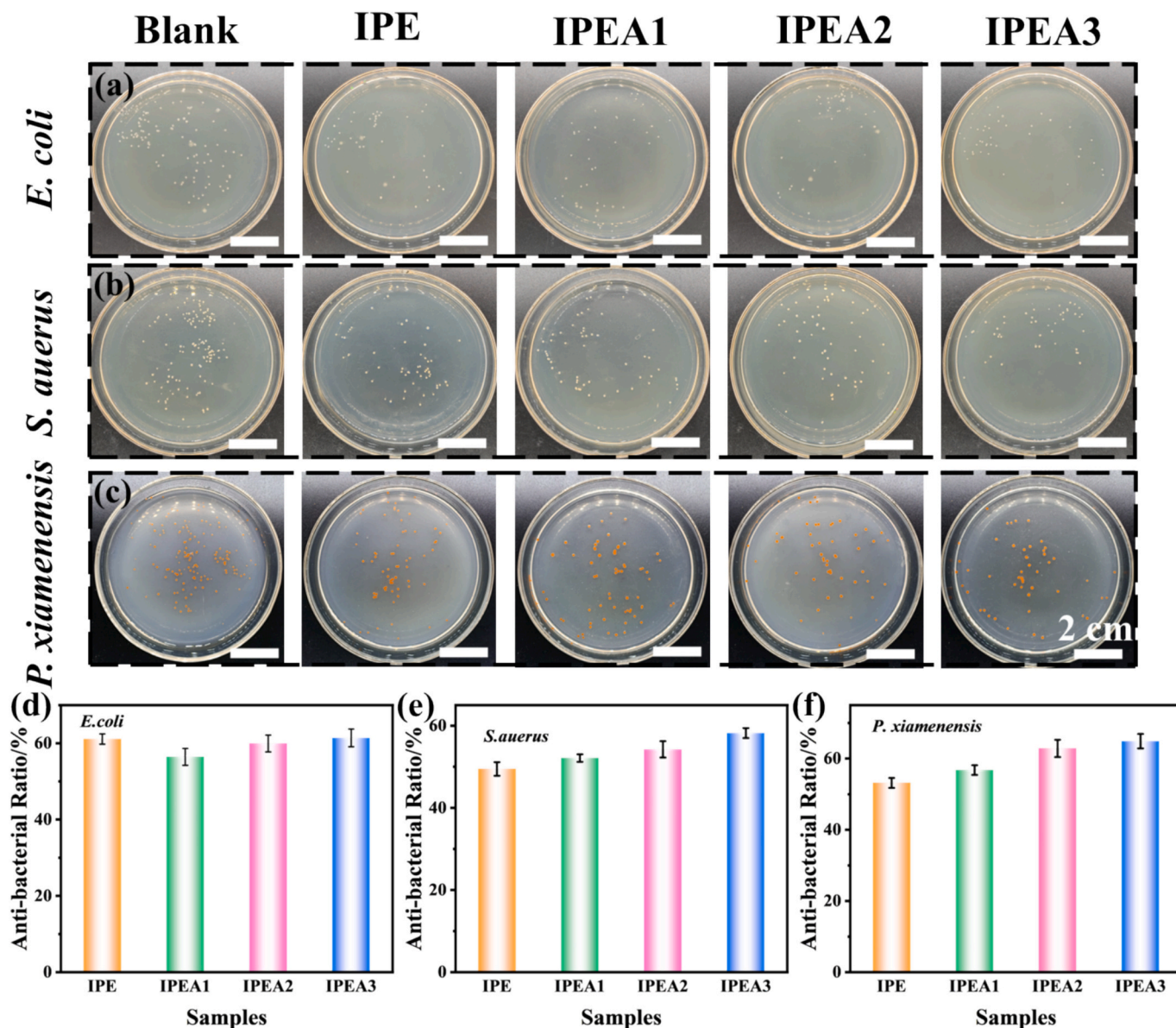


Fig. 5. Bacteria morphology of (a) *E. coli*, (b) *S. aureus* and (c) *P. xiamenensis*. Anti-bacterial ratio of coatings for (d) *E. coli*, (e) *S. aureus* and (f) *P. xiamenensis*.

were 74.2 ± 1.5 %, 93.1 ± 0.5 %, 96.6 ± 1.1 %, and 98.6 ± 0.4 %, respectively. As *N. closterium* is a type of planktonic algae that is not easy to attach to surfaces, the inhibition rate of coatings on *N. closterium* was lower than that for *Halamphora* sp. The IPEA3 coating had the best inhibition effect on the adhesion of *N. closterium*, with an inhibition rate of 74.2 ± 0.1 % after 14 days of immersion. All coatings showed good inhibition of diatom adhesion. Compared with *Halamphora* sp., the coatings had a weaker inhibitory adhesion performance to *N. closterium*, this can be ascribed to the different adhesion mechanism of diatoms.

Environmentally friendly marine antifouling coatings are of great significance for practical marine applications. Therefore, the environmental friendliness of the coatings was investigated by measuring the absorbance of algal culture solutions soaked with experimental samples. The glass plate was used as a blank sample, the IP coating (without the addition of polyethylene glycol methyl ether methacrylate) as a sample, and the IP and IPE coatings were used as control groups. The IPEAx coatings were used as the experimental group. As shown in Fig. S1, the absorbance of the algal solution gradually increased with expand the culture time, suggesting an increase in the number of diatoms. The absorbance of algal solutions immersed with different coatings showed a

consistent trend, these results indicate that the addition of polyethylene glycol methyl ether methacrylate and aminopropyl silicone oil had no influence on the growth of *N. closterium*. Therefore, the prepared coatings are environmentally friendly and have a wide application prospect. The anti-diatom effect was achieved by reducing the adhesion of diatoms, rather than by killing them.

3.5. Anti-mussel adhesion performance of coatings

The settlement of mussels on the surface of the coatings was explored. Blank samples (glass plates), and IP, IPE, and IPEAx coatings were randomly distributed on the epoxy plate, and the activity of the mussels was checked again before the experiment. Twelve mussels were selected for the experiment and randomly placed on the surface of the experimental samples. Three parallel trials were carried out, and the random arrangement of samples on the surface of the epoxy plate was changed only between the three experiments; other experimental details remained the same. The experiment was record for 72 h, and photos were taken every 12 h; the results are shown in Fig. 7. The number of mussel plaques on different surfaces was counted. The results showed

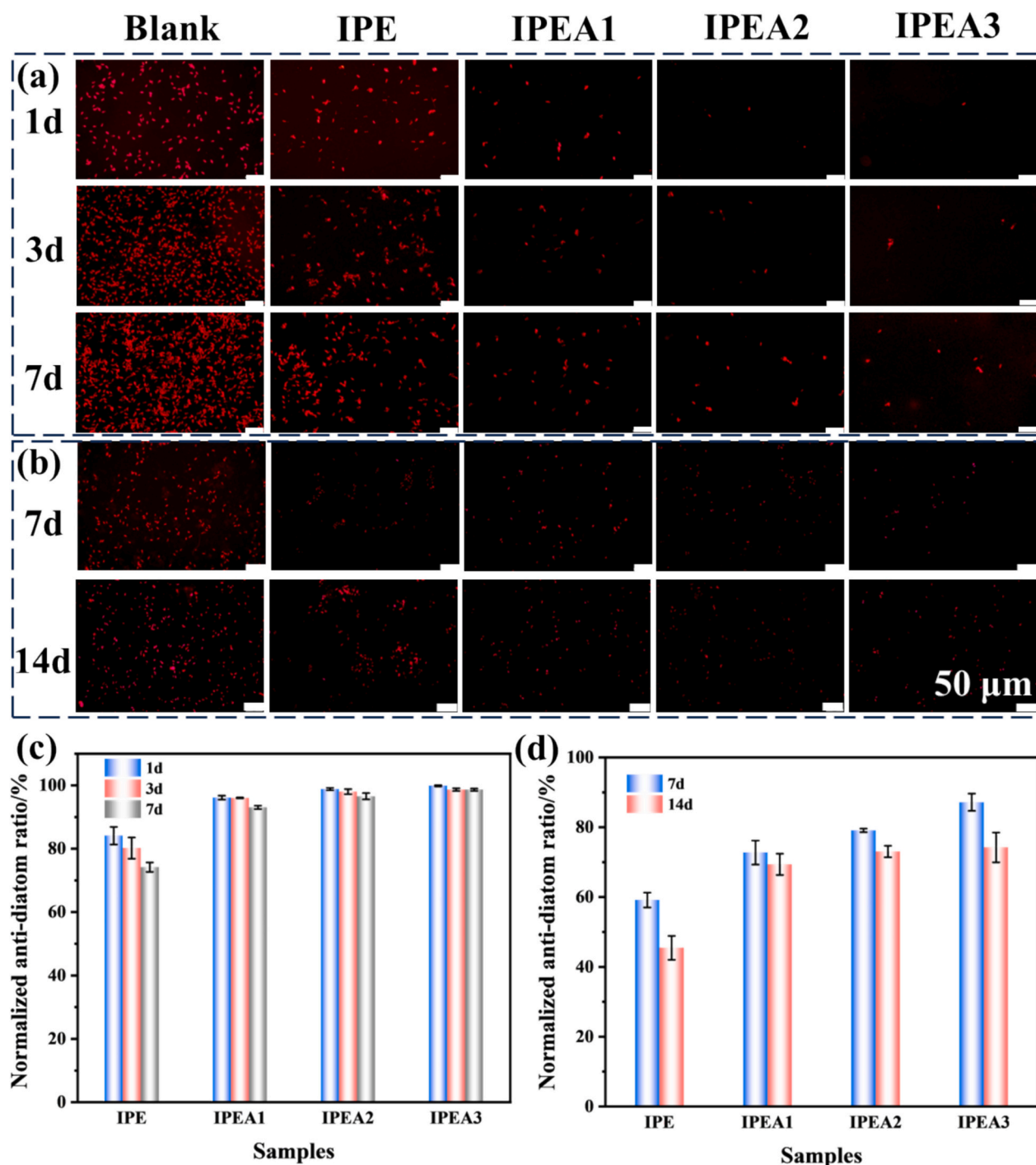


Fig. 6. The photos of (a) *Halamphora* sp. and (b) *N. closterium* attached on different coatings, anti-diatom ratio of different coatings for (c) *Halamphora* sp. and (d) *N. closterium*.

that more mussel plaques adhered to the surface of the glass sheet. Fig. 7(b) shows the percentage of adhesive plaques on different surfaces. The results showed that 36.5 ± 15.1 % of the plaques adhered to the coated surface, while 66.5 ± 10.2 % adhered to the test board and the mussels themselves, indicating that mussels were more likely to escape from the test surface and attach either to the area outside the experimental coating or to each other. Fig. 7(c) shows the photos of mussels on the coating surface every 12 h between 0 and 72 h. Fig. 7(d) shows the results of plaques adhered on the surface of each sample. It was found that IPE and IPEAx coatings can significantly inhibit the adhesion of mussels. 5.9 ± 10.2 % 2.8 ± 3.3 %, 3.2 ± 3.1 %, 3.8 ± 3.3 % of plaques were

attached on the surface of the IPE, IPEA1, IPEA2, IPEA3 coating, respectively, indicating that low surface energy coatings can significantly inhibit mussel adhesion. At the end of the experiment, the mussels were returned to the large aquarium to continue culture, and no mussels died. The good mussel activity confirmed the environmental friendliness of the experimental coatings.

4. Conclusion

Modified polyethylene glycol was prepared by a mercapto-alkene click reaction, and then modified polyethylene glycol polyurethane

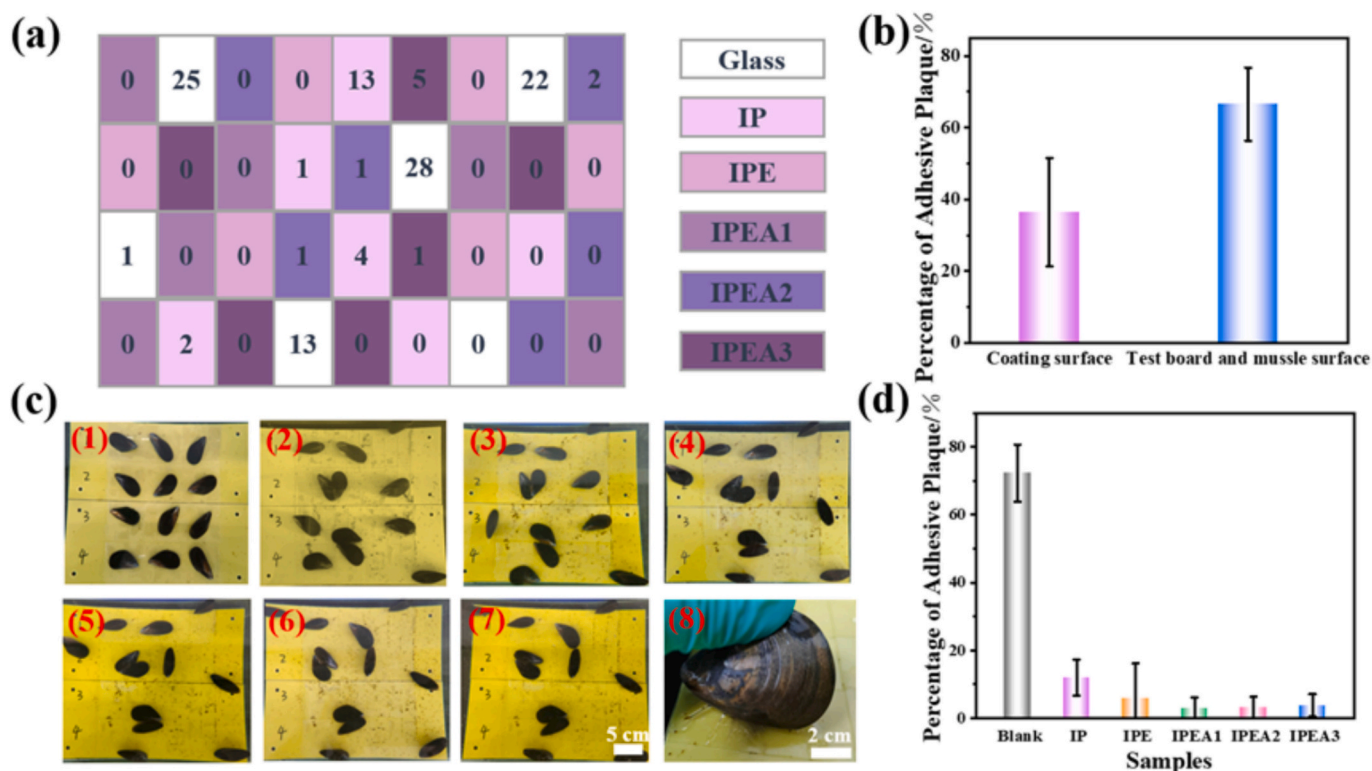


Fig. 7. (a) Samples' position and number of mussel plaques attached on the surface, (b) percentage of plaques adhered to different surface, (c) (1)–(7) were the position of mussels on the coating surface every 12 h between 0 and 72 h, (d) percentage of plaques adhered to the different coatings.

prepolymer was prepared via an addition reaction with polypropylene glycol and isophorone diisocyanate. Aminopropyl silicone oil and polyaspartate were used as chain extenders to prepare a polyurea antifouling coating. Compared with IPE, the cross-linking was intensified between aminopropyl silicone oil and polyurethane, the thermal stability, adhesion strength, tensile strength and elongation of the coating were enhanced. The antibacterial ratios of the IPEA3 coating against *E. coli*, *S. aureus* and *P. xiamenensis* were $61.4 \pm 2.3\%$, $58.2 \pm 1.2\%$, and $64.9 \pm 2.1\%$, respectively. The IPEA3 coating had the best inhibition effect of 98.6 % for *Halophora* sp., but inhibition for *N. closterium* was weaker (74.2 %). The percentage of adhesive plaques on the surface of the IPEAx coating was lower than 4 %. The results show that the coatings are effective in reducing the attachment of biofouling. After 800 rounds of friction tests, the coating still exhibited ability to inhibit diatom adhesion.

CRediT authorship contribution statement

Dandan Li: Investigation, Data curation. **Hongxia Wang:** Investigation. **Rongrong Chen:** Formal analysis, Conceptualization. **Chao Wang:** Project administration. **Hongmei Ding:** Resources. **Jun Wang:** Funding acquisition.

Declaration of competing interest

The authors declared that no conflict of interest exists for this work.

Acknowledgments

This work was supported by 2025 Beijing Postdoctoral Research Activities Funding (2025-ZZ-017), National Natural Science Foundation of China (52173199), and State Key Laboratory of Marine Coating (GZ-23-001).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109742>.

Data availability

Data will be made available on request.

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